

AKASH AJIN THOMAS

Toronto, ON

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Education

University of Toronto (St. George Campus)

Sep. 2024 – May 2029

Bachelor of Applied Science in Engineering Science + PEY Co-op

Toronto, ON

Experience

University of Toronto Robotics Association Combat Team (UTRA Combat)

Sep 2025 – Present

Student Engineer

Toronto, ON

- Redesigning “Agent P” (30 lb combat robot) into a sumo bot as part of the UTRA Combat division.
- Rebuild the robot’s control and power circuits, integrating motor drivers, sensors, and ESCs into a more effective layout
- Redesigned the chassis and weapon assembly of Agent P, optimizing for balance, center of gravity, and modularity in the 30 lb weight class.

Three Comma Capital

Jun 2025 – Aug 2025

Compliance Intern

Dubai, UAE

- Supported compliance team to review the firm’s existing Anti-Money laundering Policies and Procedures and their compliance to the Dubai Financial Services Authority rules
- Worked with a team to audit and redesign the company website to update company information and improve user interface

STER-IOT

Jul 2022 – Aug 2022

SWE Intern

Dubai, UAE

- Supported IoT engineers by developing a specialized program using Python and SQL for retrieving meter information.
- Collected critical on-site data, compiled comprehensive reports, ensuring data accuracy and actionable insights.

Projects

Attitude Control for Quadcopters | *PID controller, Python, Non-linear ODE modelling*

Sep 2025 – Present

- Model unstable flight dynamics of a quadcopter as a system of coupled, non-linear ordinary differential equations
- Design and simulate a PID feedback controller to impose stability, allowing the Unmanned Aerial Vehicle (UAV) to achieve a stable hover and reject external disturbances

Gesture-Controlled Car | *Arduino, MPU6050, NRF24L01, C++*

Oct 2025

- Designed and built a robotic car that responds to hand gestures detected via an MPU6050 gyroscope and accelerometer.
- Programmed Arduino microcontrollers to process real-time IMU data and wirelessly transmit directional commands using NRF24L01 transceivers.
- Integrated motor driver circuits to achieve smooth forward, backward, and turning motion based on wrist orientation.

Solar-Powered Car | *Chassis Design, Drivetrain Mechanics, Circuit Design*

Jun 2023

- Designed a solar powered car for a competition at De Montfort University, Dubai with custom Chassis and Drivetrain
- Placed top 5 among 35 teams.

Technical Skills

Languages: Python, C/C++, HTML/CSS, MySQL, MATLAB, SystemVerilog

Hardware: Arduino, Raspberry Pi, Circuit Design, Soldering, MPU6050, NRF24L01

Prototyping & Fabrication: Machine shop safety, lathe, mill, drill press, Onshape, Blender

Honors

UofT Engineering Dean’s List

Winter 2025

University of Toronto

UofT Engineering Dean’s List

Fall 2024

University of Toronto

Leadership Experience

Headboy, (Student Council Lead)

Apr 2023- Mar 2024

JSS private school, Dubai

- Represent the student body at assemblies, school events, and meetings with administration.
- Lead the prefect/senior student team.